

Qatar-2b

R-1532

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_200223 · created 2026-08-01T18:17:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458902.87524 +/- 0.00094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.055
Transit depth (Rp/Rs)^2	5.2 +/- 2.52 [%]
Orbital Inclination (inc)	84.08 +/- 1.74
Airmass coefficient 1 (a1)	0.38 +/- 0.15
Airmass coefficient 2 (a2)	0.489 +/- 0.064
Scatter in the residuals of the lightcurve fit is	29.87 %
Best Comparison Star	#2 - [503, 36]
Optimal Aperture	9.21
Optimal Annulus	9.98
Transit Duration (day)	0.067 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.228 $\pm$ 0.055 vs 0.16327 (deviation 0.34 σ)
Duration fit vs published	0.067 d vs 0.0767 d (deviation 0.25 σ)
Mid-time O-C	-2.3 min
Depth SNR	1.2
Residual scatter	29.87 %

Light curve

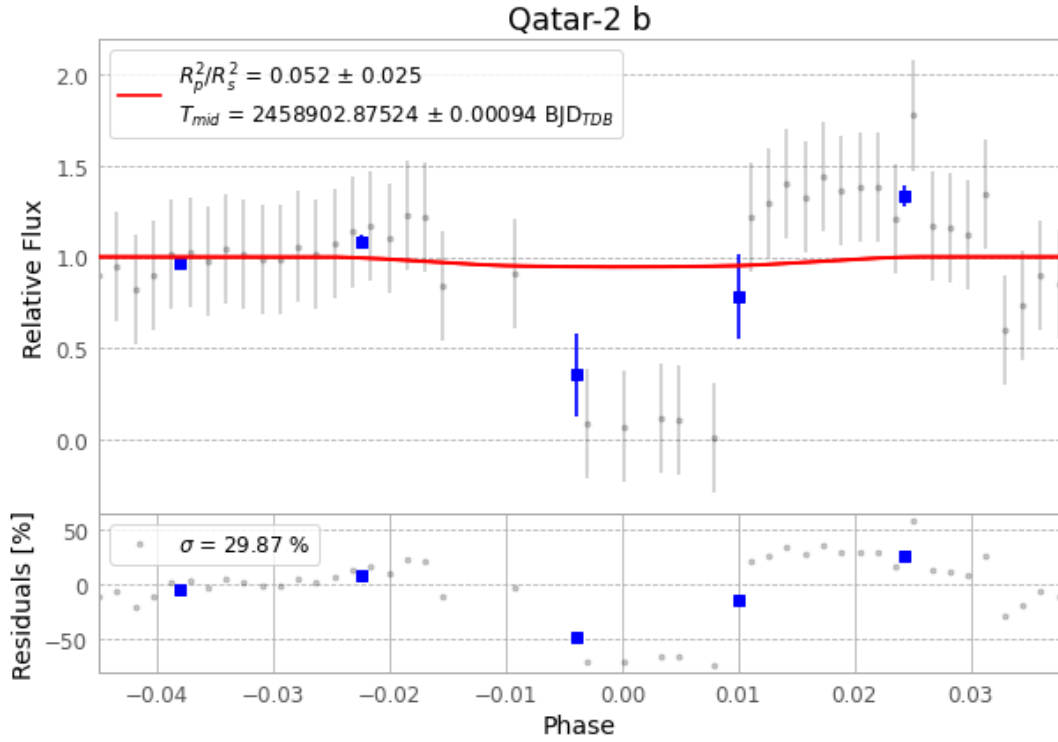


Figure 1 — FinalLightCurve_Qatar-2 b_2020-02-23.png (SHA-256 3ecb933fb2477a5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 434], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6e7ed64aeb4e3f9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar-2200223072712.FITS → Qatar-2200223110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-200223.log (c0784aaaec57...), FinalLightCurve_Qatar-2 b_2020-02-23.png (3ecb933fb247...), FinalLightCurve_Qatar-2 b_2020-02-23.csv (0cb8fbab900f...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:17Z.